A conductor decides the field before you calculate

Every conductor fact comes from one sentence: free charges move until they stop moving.

1 The four facts, each derived

A metal has free electrons. If anything pushes them, they move. Equilibrium means they have stopped, so nothing is pushing them.

WHY THE FIELD INSIDE IS ZERO

If E were non-zero inside, a free electron would feel eE and move. It has stopped, so $E_{\text{inside}} = 0$.

WHY THE FIELD MUST BE PERPENDICULAR TO THE SURFACE

On the surface the potential is constant. Along the surface $dV = 0$, and $dV = -\vec{E} \cdot d\vec{l}$ for every such $d\vec{l}$: only the normal component survives.

HOW LARGE THE SURFACE FIELD IS (PILLBOX)

Take a thin pillbox straddling the surface, face area A, only the outside face is in a field: $EA = \sigma A/\varepsilon_0$.
$\Rightarrow \; E_{\text{surface}} = \sigma/\varepsilon_0$

learn these four

$E_{\text{inside}} = 0$; the whole conductor, cavity walls included, is one potential
just outside: $E = \sigma/\varepsilon_0$, perpendicular to the metal
a cavity holding $+q$ shows $-q$ on its wall and $+q$ on the outer surface
an empty cavity is shielded from fields outside it

TRAP

$E = 0$ is not $V = 0$. A charged shell has no field inside and still sits at $V = kQ/R$: zero slope, non-zero height.

2 Cavities: where the charge goes

RESULT

INDUCED CHARGE, FROM ONE GAUSSIAN SURFACE

Draw any closed surface inside the metal. Every point on it has $E = 0$, so $\oint \vec{E} \cdot d\vec{A} = 0$.

The enclosed charge must then be zero: the wall carries $-q$, and since the conductor is neutral, $+q$ appears outside.
The outer surface knows only the total charge, not where it is hidden - so a cavity is shielded.

3 Gauss in three steps

Name a surface where $|\vec{E}|$ is constant and $\vec{E}$ is parallel or perpendicular to $d\vec{A}$, then use $\oint \vec{E} \cdot d\vec{A} = Q_{\text{enc}}/\varepsilon_0$.

The flux integral is only useful after symmetry has turned it into E times an area. Otherwise Gauss is still true, just not calculable.

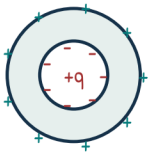
SOURCE	FIELD AT DISTANCE R	NOTE
point charge, outside a sphere	kQ/r^2	the starting point
inside a uniform solid sphere	$\rho r/3\varepsilon_0$	zero at the centre
long line, λ per length	$\lambda/2\pi\varepsilon_0 r$	falls as $1/r$
large sheet, σ per face	$\sigma/2\varepsilon_0$	away on both sides
conductor surface	σ/ε_0	one side only
infinite slab, half-width a	$\rho x/\varepsilon_0$	outside: $\rho a/\varepsilon_0$

one line of numbers

Radius $10\ \text{cm}$, charge $1\ \mu\text{C}$: $V_s = 9.0 \times 10^4\ \text{V}$, $E_s = 9.0 \times 10^5\ \text{V/m}$; both fall off outside as $1/r$ and $1/r^2$.

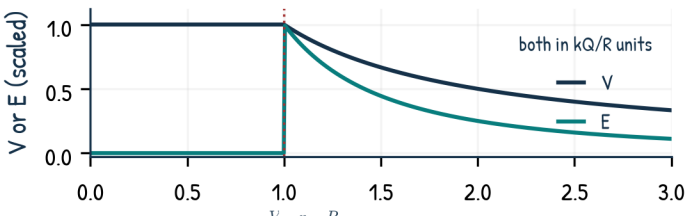
TRAP

One charged plate gives $\sigma/2\varepsilon_0$, but a capacitor gap gives σ/ε_0 - two plates, two contributions.



no field in the metal, so the wall must cancel +q exactly; the outside then sees only the total charge of the shell.

A neutral shell with $+q$ in the cavity: $-q$ on the wall, $+q$ outside, $E = 0$ in the metal.



Field is flat and non-zero inside; the step in V at $r = R$ marks where the charge sits.

REMEMBER

Say where the charge sits, then what the potential is. Neither step needs a capacitor formula.

Capacitance: the plate formula and the one slab rule

Capacitance is charge needed per volt. Derive it once for plates, then a single rule covers dielectric, metal and vacuum.

1 Why C depends on geometry alone

THE RATIO IS FIXED BY SHAPE AND MEDIUM

Every field and potential is proportional to the charges that produce it. Double every q and every V doubles. So Q/V cannot depend on Q. What is left is geometry and the medium: $C = \varepsilon_0 A/d$ for plates, $4\pi\varepsilon_0 R$ for an isolated sphere.

TRAP

A capacitor holds $+Q$ and $-Q$, so its net charge is zero. "Charge on the capacitor" always means the positive plate.

2 The plate formula, derived in three lines

RESULT

THE PLATE FORMULA

Gap field of two opposite plates: $E = \dfrac{Q/2}{\varepsilon_0 A} + \dfrac{Q/2}{\varepsilon_0 A} = \dfrac{Q}{\varepsilon_0 A}$ (the two plates add).

Voltage between them: $V = Ed = \dfrac{Qd}{\varepsilon_0 A}$.

Divide: $C = \dfrac{Q}{V} = \dfrac{\varepsilon_0 A}{d}$, and with a dielectric filling the gap $C = \dfrac{K\varepsilon_0 A}{d}$.

the two you quote

$C = \dfrac{\varepsilon_0 A}{d}$, $C = \dfrac{K\varepsilon_0 A}{d}$; $\varepsilon_0 = 8.85 \times 10^{-12}\ \mathrm{F/m}$, $4\pi\varepsilon_0 = 1.11 \times 10^{-10}\ \mathrm{F/m}$

two plates, two answers

Area $100\ \mathrm{cm}^2$, gap $1\ \mathrm{mm}$: air $88.5\ \mathrm{pF}$; with $K = 5$, $443\ \mathrm{pF}$. Doubling the gap halves C; an isolated $1\ \mathrm{cm}$ sphere is only $1.1\ \mathrm{pF}$.

3 One slab rule for every gap

RESULT

WHY THE THICKNESS BECOMES T/K

With a slab of thickness t and constant free charge: $E_{\mathrm{vac}} = \dfrac{Q}{\varepsilon_0 A}$, $E_{\mathrm{slab}} = \dfrac{Q}{K\varepsilon_0 A}$.

Add the two voltage drops: $V = E_{\mathrm{vac}}(d - t) + E_{\mathrm{slab}}\, t = \dfrac{Q}{\varepsilon_0 A}\left(d - t + \dfrac{t}{K}\right)$.

Divide by Q: $C = \dfrac{\varepsilon_0 A}{d - t + t/K}$. Metal is the limit $K \to \infty$, giving $C = \dfrac{\varepsilon_0 A}{d - t}$ - the thickness is simply removed.

use it in all three cases

dielectric slab: $C = \dfrac{\varepsilon_0 A}{d - t + t/K}$; metal slab: $C = \dfrac{\varepsilon_0 A}{d - t}$; vacuum: $C = \dfrac{\varepsilon_0 A}{d}$

two layers: $\dfrac{1}{C} = \dfrac{t_1}{K_1 \varepsilon_0 A} + \dfrac{t_2}{K_2 \varepsilon_0 A}$; only total thickness of each layer matters

TRAP

Use the overlap area, never the sum of both plate areas. And $C = \varepsilon_0 A/d$ needs d small compared with the plate width; edge fringing always adds a little.

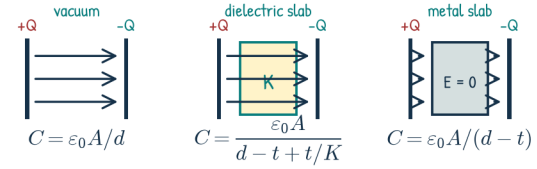
4 Many plates, and plates with unequal charge

N PLATES AND TWO ODD CHARGES

Interleave n plates, alternate ones wired to the two terminals: there are $n - 1$ gaps, all at the same voltage, so $C = (n-1)\varepsilon_0 A/d$.

Large plates carrying Q_1, Q_2: the facing surfaces hold $\pm(Q_1 - Q_2)/2$, since only that much can be shared across the gap.

Then $V_1 - V_2 = \dfrac{(Q_1 - Q_2)d}{2\varepsilon_0 A}$. A net plate charge is not the same as the facing charge.



$C = \varepsilon_0 A/d$ $\quad C = \dfrac{\varepsilon_0 A}{d - t + t/K}$ $\quad C = \varepsilon_0 A/(d - t)$

Every gap is $C = \varepsilon_0 A/(d - t + t/K)$; metal is $K \to \infty$.



A floating metal slab removes its own thickness from the voltage drop.

REMEMBER

C is geometry plus medium. Derive the plate formula once, then replace the filled thickness by t/K.

Sphere, cylinder and connected conductors

One method: assign the charge, find E by Gauss, integrate to get V, divide. The geometry only changes the integral.

1 Potential inside, and why the graph is flat

POTENTIAL OF A CHARGED SHELL

Outside, $V(r) = -\int_{\infty}^{r} kQ/r'^2\,dr' = kQ/r$.

Inside, the path in from the surface costs nothing because $E = 0$, so $V = kQ/R$ all the way in. The graph steps at $r = R$ because that is where the charge sits, not because V jumps.

2 Concentric spheres

RESULT

THE SPHERICAL CAPACITOR

Gauss between the spheres: $E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$, independent of the outer shell.

Integrate from a to b: $V = \dfrac{Q}{4\pi\varepsilon_0}\left(\dfrac{1}{a} - \dfrac{1}{b}\right)$.

Divide: $C = 4\pi\varepsilon_0 \dfrac{ab}{b-a}$. Let $b \to \infty$: the isolated sphere $4\pi\varepsilon_0 a$.

Let $b - a \ll a$: the area is $4\pi a^2$ and the plate formula returns - always check a new geometry against that limit.

3 Coaxial cylinders

THE CYLINDRICAL CAPACITOR

Gauss on a cylinder of radius r and length L: $E \cdot 2\pi r L = \dfrac{\lambda L}{\varepsilon_0}$, so $E = \dfrac{\lambda}{2\pi\varepsilon_0 r}$.

Integrate $1/r$: $V = \dfrac{\lambda}{2\pi\varepsilon_0}\ln\dfrac{b}{a}$, hence $C = \dfrac{2\pi\varepsilon_0 L}{\ln(b/a)}$.

Radial layers in series add their logs: $\dfrac{1}{C/L} = \sum_i \dfrac{\ln(r_{i+1}/r_i)}{2\pi\varepsilon_i}$.

4 Two connected spheres: potential is shared, charge is not

WHY THE SMALLER SPHERE WINS THE FIELD

A wire makes one conductor, so $V_a = V_b$: $\dfrac{Q_a}{a} = \dfrac{Q_b}{b} \Rightarrow Q_a = Q\dfrac{a}{a+b}$.

Surface fields: $E = kQ_a/a^2$ and $E = kQ_b/b^2$, so $\dfrac{E_a}{E_b} = \dfrac{b}{a}$ - the smaller sphere has the larger field.

That is why a pointed conductor sparks first.

TRAP

These distant-sphere results need separation $\gg$ radius. For nearby spheres the induced charge makes the surface distribution non-uniform and V is no longer kQ/r.

5 Two more geometries worth having

RESULT

results worth quoting

two distant spheres, radii a, b, opposite charges: $C \simeq 4\pi\varepsilon_0 \dfrac{ab}{a+b}$ (this is the series combination $1/C = 1/C_a + 1/C_b$)

sphere of radius a coated to radius b with dielectric K: $\dfrac{1}{C} = \dfrac{(1/a - 1/b)/K + 1/b}{4\pi\varepsilon_0}$ - add the dielectric drop to the vacuum drop

numbers that catch mistakes

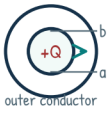
Coax with $a = 2$ mm, $b = 10$ mm, $L = 1$ m: $C = 2\pi\varepsilon_0/\ln 5 = 77$ pF; the same cable in oil of $K = 2.2$ gives 170 pF.

Two spheres 1 cm and 4 cm sharing $100\ \mu$C: the small one takes $20\ \mu$C but has $16\times$ the surface field of the large one.



field lives only between a and b:
$E = kQ/r^2$
zero inside;
kQ/r^2 outside

Concentric spheres: $E \propto 1/r^2$, and only the part between a and b sets V.



field lives only between a and b:
$E = \lambda/2\pi\varepsilon_0 r$
zero inside and outside;
length L cancels in C

Coaxial cylinders: $E \propto 1/r$, so the voltage brings a logarithm.

REMEMBER

Assign Q, get E from Gauss, integrate to V, divide. Check every new geometry against the plate limit.

Networks: islands, bridges, ladders

Wires make islands. Charge is conserved on each island, not on each capacitor - derive every combination from that.

1 Series and parallel, derived

SERIES: ONE OVER C ADDS

Two capacitors in a line make the middle island. If it started neutral it stays neutral, so both carry the same Q.
The voltages add: $V = \frac{Q}{C_1} + \frac{Q}{C_2}$, so $\frac{1}{C_{\text{eq}}} = \frac{1}{C_1} + \frac{1}{C_2}$.
The smallest capacitor then takes the largest share of the voltage.

PARALLEL: C ADDS

Both are across the same two nodes, so V is common and the charges add: $Q = C_1 V + C_2 V$.
$\Rightarrow C_{\text{eq}} = C_1 + C_2$. Energy follows: series $\frac{U_i}{U} = \frac{C_{\text{eq}}}{C_i}$, parallel $\frac{U_i}{U} = \frac{C_i}{C_{\text{eq}}}$.

TRAP

Series or parallel is decided by the wires, not by the drawing - redraw first. A precharged middle island also breaks the equal-charge rule.

2 The island method that never fails

METHOD

ONE CONSERVATION LAW PER CONDUCTOR

Each floating island has a fixed net charge Q_i^{init}. Every plate on it holds $C(V_i - V_j)$ against a neighbour.
Add them with signs: $\sum_j C_{ij}(V_i - V_j) = Q_i^{\text{init}}$. One equation per floating island.
Solve for the potentials, then read each capacitor as $q = C\,\Delta V$.

Island types: grounded $\Rightarrow V = 0$; battery-connected $\Rightarrow$ difference fixed; floating $\Rightarrow$ net charge fixed.

3 Bridge, cube, ladder, star-delta

RESULT

BALANCED BRIDGE

Each arm divides the same applied voltage: $V_X = V\frac{C_1}{C_1 + C_2}$ and $V_Y = V\frac{C_3}{C_3 + C_4}$.
It carries nothing when $V_X = V_Y$, i.e. $\frac{C_1}{C_2} = \frac{C_3}{C_4}$; otherwise write two island equations.

INFINITE LADDER

One more repeat unit must change nothing: the remainder X is in series with C, then in parallel with C.
$X = C + \frac{CX}{C + X} \Rightarrow X = \frac{\sqrt{5} - 1}{2} C$ (the negative root is rejected: $C > 0$).

results worth memorising

cube of twelve equal C: opposite vertices $\frac{6C}{5}$, edge $\frac{12C}{7}$, face diagonal $\frac{4C}{3}$
star-delta with $S = ab + bc + ca$: arms $x = S/b$, $y = S/c$, $z = S/a$; back again $C_{12} = xy/(x + y + z)$

TRAP

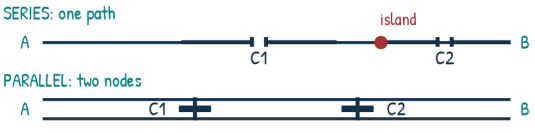
Do not copy the resistor star-delta across: for capacitors the star arm is a sum of products divided by one branch.

reduce, then audit the pieces

$6\ \mu\text{F}$ and $3\ \mu\text{F}$ in series give $2\ \mu\text{F}$; with $4\ \mu\text{F}$ in parallel, $C_{\text{eq}} = 6\ \mu\text{F}$. At $12\ \text{V}$: $Q = 72\ \mu\text{C}$, $U = 432\ \mu\text{J}$.
Audit: the series pair holds $24\ \mu\text{C}$ at $4\ \text{V}$ and $8\ \text{V}$, the $4\ \mu\text{F}$ holds $48\ \mu\text{C}$; and $24 + 48 = 72$.

4 Charge through a wire

Cut the circuit with a surface crossed by that wire only. The charge that crossed equals the change in enclosed plate charge, signs included - it is not $C\Delta V$ in general.



The middle island is what forces series charges to be equal.



Balanced means $V_X = V_Y$: delete the bridge, or write one island equation at each node.

REMEMBER

Name the terminals, mark the islands, write one charge equation per island. Folding is a shortcut that must be justified.

Energy, the battery, and the heat bill

Every energy result is an integral or a ledger. The half comes from the triangle, and the battery is part of the accounting.

1 Stored energy, integrated

RESULT

WHERE THE HALF COMES FROM

Push a small charge dq in while the voltage is $V = q/C$: the work is $dU = V\,dq = \frac{q}{C}dq$.

Integrate from 0 to Q: $U = \frac{Q^2}{2C} = \frac{1}{2}QV = \frac{1}{2}CV^2$.

On a V–q graph this is the area of a triangle, not a rectangle – that is the missing half.

FIELD ENERGY DENSITY

For a parallel plate: $U = \frac{1}{2}CV^2 = \frac{1}{2}\cdot\frac{\varepsilon_0 A}{d}\cdot(Ed)^2 = \frac{1}{2}\varepsilon_0 E^2\,(Ad)$.

Divide by the volume Ad: $u = \frac{1}{2}\varepsilon_0 E^2$ per unit volume, wherever the field is.

the check that catches sign errors

$2\ \mu\text{F}$ at $100\ \text{V}$: $Q = 200\ \mu\text{C}$, $U = 0.1\ \text{J}$. Pull the plates to twice the gap with the battery off: C halves, Q stays, so V and U double – you supplied that work.

2 The one fork: fixed Q or fixed V

TABLE

WHY THE TWO COLUMNS DIFFER

At fixed Q: $C \to KC$ and Q cannot change, so $V = Q/C$ falls by K and $U = Q^2/2C$ falls by K. Energy leaves into the mechanical work.
At fixed V: V is pinned, so $Q = CV$ grows by K and $U = \frac{1}{2}CV^2$ grows by K. The cell supplies both the extra energy and the mechanical work.

AFTER FULL DIELECTRIC INSERTION	FIXED Q (BATTERY OFF)	FIXED V (BATTERY ON)
C	grows by K	grows by K
V	falls by K	unchanged
E in the gap	falls by K	unchanged, V/d
Q	unchanged	grows by K
U	falls by K	grows by K
battery work	zero	$V^2\Delta C$

the fork in numbers

$1\ \mu\text{F}$ at $100\ \text{V}$, then $K = 2$ fills the gap. Battery on: C becomes $2\ \mu\text{F}$, the cell supplies $V^2\Delta C = 10\ \text{mJ}$, U rises by $5\ \text{mJ}$, and the other $5\ \text{mJ}$ goes into pulling the slab in.
Battery off: $Q = 100\ \mu\text{C}$ is fixed, so V drops to $50\ \text{V}$ and U falls from $5\ \text{mJ}$ to $2.5\ \text{mJ}$ – that released energy is the work the field does on the slab.

TRAP

Battery work is not field energy. The cell supplies $V^2\Delta C$ while U rises by only $\frac{1}{2}V^2\Delta C$; differentiating U alone gives the wrong sign for the force.

3 The heat bill when two capacitors are joined

CHARGE CONSERVATION FIRST, ENERGY SECOND

Islands conserve charge: $C_1V_1 + C_2V_2 = (C_1 + C_2)V_f$, so $V_f = \dfrac{C_1V_1 + C_2V_2}{C_1 + C_2}$.

Energy before minus after: $H = \dfrac{C_1C_2}{2(C_1 + C_2)}(V_1 - V_2)^2$ – never negative, and independent of the resistance.

numbers on the heat bill

$C_1 = 2\ \mu\text{F}$ at $100\ \text{V}$, $C_2 = 3\ \mu\text{F}$ uncharged: $V_f = 40\ \text{V}$ and $H = \frac{1}{2}\cdot\frac{6}{5}\ \mu\text{F}\cdot 100^2 = 6\ \text{mJ}$.
Check: $10\ \text{mJ}$ before, $4\ \text{mJ}$ after – the missing $6\ \text{mJ}$ is heat. Charging an empty C from a cell loses exactly half of CE^2.



Work is the area under the V–q line: a triangle, hence $\frac{1}{2}$.

What is held fixed?

Isolated	Battery connected
Q cannot change	V is pinned
minimise U	minimise U – VQ
$F = -dU/dx$	$F = -d\Pi/dx$

Ask what is held fixed before differentiating. It is the whole chapter in one picture.

REMEMBER

Fixed Q: differentiate U. Fixed V: differentiate $U - QV$. Heat: conserve charge, then subtract energies.

Force on plates and dielectrics, from energy

Write C(x), state the constraint, differentiate. The factor one-half is the thing examiners test.

1 Force from energy, and the pressure of a charged sphere

FORCE FROM THE DERIVATIVE

At fixed Q: $F = -\dfrac{d}{dx}\left(\dfrac{Q^2}{2C}\right) = \dfrac{Q^2}{2C^2}\dfrac{dC}{dx}$.

At fixed V the battery must be included, $\Pi = U - QV = -\tfrac{1}{2}CV^2$, so $F = -\dfrac{d\Pi}{dx} = \tfrac{1}{2}V^2\dfrac{dC}{dx}$.

Same state $Q = CV$ gives the same force either way - the constraint only decides what happens next.

OUTWARD PRESSURE ON A CHARGED SPHERE

Expand the radius by dR at fixed Q: $U = \dfrac{Q^2}{8\pi\varepsilon_0 R}$ falls, and $dU = -p\,4\pi R^2 dR$.

$\Rightarrow p = \dfrac{Q^2}{32\pi^2\varepsilon_0 R^4}$, pushing outward; the net force on the sphere is still zero by symmetry.

force between real plates

$100\ \text{cm}^2$ at $1\ \text{mm}$, $100\ \text{V}$: $p = \tfrac{1}{2}\varepsilon_0 E^2 = 0.044\ \text{Pa}$, $F = 4.4\times10^{-4}\ \text{N}$.

Halve the gap at fixed Q: E is unchanged, so F is unchanged. At fixed V: E doubles, F quadruples.

2 The force between the plates

RESULT

WHY IT IS HALF THE GAP FIELD

A plate does not pull on itself. The field at one plate due to the other is $\dfrac{Q}{2\varepsilon_0 A}$ - half the gap field.

So $F = Q\cdot\dfrac{Q}{2\varepsilon_0 A} = \dfrac{Q^2}{2\varepsilon_0 A}$; as a pressure, $p = \dfrac{\sigma^2}{2\varepsilon_0}$.

The same answer from energy: $F = -\dfrac{dU}{dx}$ with $U = \dfrac{Q^2 x}{2\varepsilon_0 A}$ gives $F = -\dfrac{Q^2}{2\varepsilon_0 A}$, inward.

TRAP

Do not write $F = QE$ with the full gap field σ/ε_0; the correct force is $QE/2$. That half is the whole question.

3 A dielectric is always pulled inward

RESULT

SLAB ENTERING BY X

Filled and empty parts are in parallel: $C(x) = \dfrac{\varepsilon_0 w}{d}[L + (K-1)x]$, so $\dfrac{dC}{dx} = \dfrac{\varepsilon_0(K-1)w}{d}$.

At fixed V: $F = \tfrac{1}{2}V^2\dfrac{dC}{dx} = \dfrac{\varepsilon_0(K-1)w}{2d}V^2$, constant, pulling the slab in.

At fixed Q: use $V = Q/C(x)$, so $F = \dfrac{Q^2}{2C^2}\dfrac{dC}{dx}$ - same direction, smaller as the slab goes deeper.

LIQUID CLIMBING BETWEEN PLATES

Electric force per unit plate width $= \dfrac{\varepsilon_0(K-1)V^2}{2d}$; weight of the raised column per width $= \rho g d h$.

Balance: $h = \dfrac{\varepsilon_0(K-1)V^2}{2\rho g d^2}$ - linear in $K-1$, quadratic in V, falling fast as the gap opens.

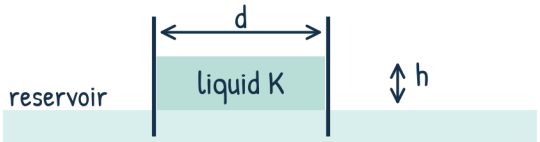
the three force results

plate: $F = \dfrac{Q^2}{2\varepsilon_0 A} = \dfrac{\varepsilon_0 A V^2}{2d^2}$; slab: $\dfrac{\varepsilon_0(K-1)w}{2d}V^2$; liquid: $h = \dfrac{\varepsilon_0(K-1)V^2}{2\rho g d^2}$

all three point toward larger C - the one thing that never changes



Filled and empty parts are parallel capacitors, so $dC/dx > 0$: the slab is dragged in.



The same force lifts a liquid: electric work pays for the rise in gravitational energy.

REMEMBER

Write $C(x)$, name the constraint, differentiate the right energy. Force always points toward larger C.

Image charge, and a plate that snaps down

Two questions that look hard and are one construction each: a mirror charge, and an equilibrium that stops existing.

1 Image charge: the one extension worth keeping

KEPT

A grounded plane is an equipotential, and two opposite charges of equal size make a whole plane equipotential - so the plane can be replaced by the second charge.

FORCE, ENERGY AND INDUCED CHARGE

Replace the plane by $-q$ at depth h. The real charge is then $2h$ from its image: $F = \dfrac{q^2}{4\pi\varepsilon_0(2h)^2} = \dfrac{q^2}{16\pi\varepsilon_0 h^2}$.

Bringing the charge in from infinity: $W = \frac{1}{2}q\,(V_{\text{image}}) = \dfrac{q^2}{16\pi\varepsilon_0 h}$ - the half is because the induced charge builds up as you approach.

The plane carries induced charge $-q$, spread with $\sigma(\rho) = -\dfrac{qh}{2\pi(\rho^2 + h^2)^{3/2}}$, largest directly below the charge.

TRAP

Using $q^2/8\pi\varepsilon_0 h$ double counts the work of building the induced charge. The energy is half of $q\times$ the image potential, not of $q\times$ the full pair energy.

Why a mirror charge is allowed at all: in the region above the plane, solving Laplace with the plane at $V = 0$ gives a unique field. The pair of charges already satisfies that boundary condition, so it must be the right field. Above the plane nobody can tell the difference; below it there is no field to speak of, and the metal is not the image.

image results for a grounded plane

force $\dfrac{q^2}{16\pi\varepsilon_0 h^2}$, work to bring from infinity $\dfrac{q^2}{16\pi\varepsilon_0 h}$, induced charge $-q$

surface density $\sigma(\rho) = -\dfrac{qh}{2\pi(\rho^2 + h^2)^{3/2}}$: sharpest directly under the charge, and it integrates to $-q$

grounded sphere of radius a (optional, one line): image $q' = -qa/d$ at a^2/d; an isolated neutral sphere needs a second image $+qa/d$ at the centre

2 Where the method stops

3 Stretch: a plate held on a spring

EQUILIBRIUM, AND WHERE IT BREAKS DOWN

The gap is $d - x$ and the spring pulls back with kx, so $kx = \dfrac{\varepsilon_0 A V^2}{2(d-x)^2}$. Write $y = \dfrac{\varepsilon_0 A V^2}{2kd^2}$ and solve $x(1 - x/d)^2 = yd$.

The two roots merge when the derivative vanishes too: $x = d/3$, at $V^2 = \dfrac{8kd^3}{27\varepsilon_0 A}$. Beyond it no equilibrium exists and the plate snaps down.

At fixed charge the force $Q^2/2\varepsilon_0 A$ does not depend on x, so there is no fold and no snap-down.

4 Stretch: a plate held on a spring

EQUILIBRIUM, AND WHERE IT BREAKS DOWN

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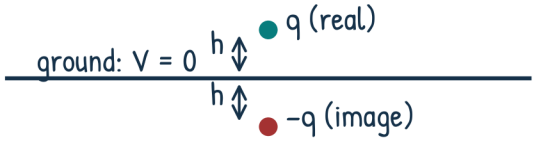
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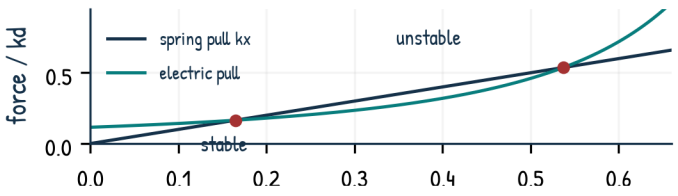
numbers for the image result

$q = 1\ \mu\text{C}$ at $h = 1\ \text{cm}$ above a grounded plane: $F = 0.225\ \text{N}$ and the work to bring it from infinity is $0.45\ \text{mJ}$.

At $2\ \text{cm}$ the force is four times smaller and the energy half: both scale as $1/h^2$ and $1/h$, so the energy is always the force times $h/2$.



The image makes the plane equipotential; real-image separation is $2h$, not h.



Two equilibria appear, then merge and disappear: that is the snap-down.

REMEMBER

Mirror the charge to make the boundary equipotential, then treat it as a real Coulomb problem - but count energy once, not twice.

Dielectrics: why the field falls, and layers vs patches

Polarisation puts bound charge on the faces. Whether the field falls is decided by the constraint, not by the material alone.

1 What the material actually does

WHY E FALLS BY K AT FIXED FREE CHARGE

The applied field lines up molecular dipoles; their bound charge appears on the faces, opposing the free charge.
The net field inside is the free field minus the bound contribution: $E = \frac{\sigma_{\text{free}} - \sigma_b}{\varepsilon_0} = \frac{\sigma_{\text{free}}}{K\varepsilon_0}$, using $\sigma_b = \sigma_{\text{free}}\left(1 - \frac{1}{K}\right)$.
That is the whole of "the dielectric reduces the field": it happens because the bound charge is there, not because the material absorbs field.

WHY NOTHING FALLS WITH THE BATTERY ON

The cell holds the plates at fixed V, so $E = V/d$ cannot change. Instead it pushes in more free charge until $Q = KQ_0$.
Capacitance rises by K either way - it is the charge, not the field, that does the extra work.

one-line summary

fixed Q: $E = \frac{E_0}{K}$, $V = \frac{V_0}{K}$, $C = KC_0$, $U = \frac{U_0}{K}$ | fixed V: $E = E_0$, $C = KC_0$, $Q = KQ_0$, $U = KU_0$

TRAP

"A dielectric always reduces the field" is false. Fill the gap with the battery still connected and E is untouched - the charge grows instead.

2 Layers or patches: two derivations

RESULT

LAYERS ALONG THE FIELD (SERIES)

No free charge at the interface, so the same $D = Q/A$ crosses both layers: $E_i = \frac{Q}{K_i\varepsilon_0 A}$.
Voltages add: $V = \sum \frac{Q\,t_i}{K_i\varepsilon_0 A} \Rightarrow \frac{1}{C} = \sum \frac{t_i}{K_i\varepsilon_0 A}$, and for one slab $C = \frac{\varepsilon_0 A}{d - t + t/K}$.
Half the thickness filled: $C = \frac{2KC_0}{K+1}$; as $K \to \infty$ the metal-slab answer returns.

PATCHES ACROSS THE AREA (PARALLEL)

Every patch sees the same V and the same $E = V/d$, but each carries its own charge: $Q_i = \frac{K_i\varepsilon_0 A_i}{d}\Delta V$.
Sums add: $C = \sum \frac{K_i\varepsilon_0 A_i}{d}$; half the area filled gives $C = C_0\frac{K+1}{2}$.

TRAP

Which case is which? Cut the plates into pieces and you have patches. Cut across the gap and you have layers. Layers add voltages, patches add charges.

3 Dielectric strength sets the design

RATING

RATINGS AND WHY SERIES IS DANGEROUS

Breakdown needs E_b, so the safe voltage is $V_{\max} = E_b d$; the best stored density is $\frac{1}{2}\varepsilon_0 K E_b^2$. Air: 3×10^6 V/m, so 0.1 mm holds about 300 V.
In series, equal charge means $V_i \propto 1/C_i$: the smallest capacitor takes the largest voltage and fails first, so a chain is limited by its weakest member.

numbers worth checking twice

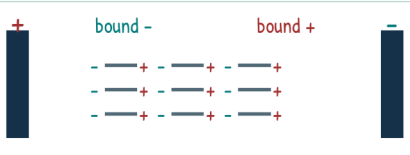
A $2\ \mu$F air capacitor, half the gap filled with $K = 4$: layers give $\frac{2 \cdot 4}{5}C_0 = 1.6C_0$, patches give $\frac{4+1}{2}C_0 = 2.5C_0$. Different answers, same picture - the geometry decides.

4 Six checks before you hand in

Dimensions: an isolated body has $C \sim \varepsilon_0 \times$ length. Limits: $K \to 1$ returns vacuum, $b \to \infty$ returns the isolated sphere. Signs: heat cannot be negative, and a dielectric is always pulled toward larger C. Endpoints: a transient starts and ends at values the circuit allows. Stability: check curvature when equilibrium is asked. Ratings: compare each local field with E_b.



Layers share the normal D and add voltages; patches share the voltage and add charges.



Bound charge on the faces opposes the free charge: that is the whole mechanism.

REMEMBER

Fixed Q the field falls by K; fixed V it does not. Layers add voltages, patches add charges.

RC transients solved, and the capacitor in AC

One differential equation gives every transient. Two switch rules answer the rest, and one derivative gives the AC response.

1 Solve the circuit once

METHOD

WHY EVERY ANSWER IS AN EXPONENTIAL

A cell E, resistor R, capacitor C in series: $E = Ri + V_C$ with $i = C\frac{dV_C}{dt}$, so $RC\frac{dV_C}{dt} = E - V_C$.

Separate and integrate: $V_C(t) = E + (V_0 - E)e^{-t/RC}$. The approach to the final value is exponential because the drive is proportional to the remaining gap.

Any linear network reduces to this with V_∞ and R_{th}: $V(t) = V_\infty + (V_0 - V_\infty)e^{-t/\tau}$, $\tau = R_{\text{th}}C$.

the three numbers

V_0 : capacitor voltage just before switching - it cannot jump

V_∞ : remove the capacitor (open circuit) and solve the resistors

R_{th} : resistance seen from the capacitor terminals, sources killed, dependent sources kept

Clock: one time constant 63 %, two 86 %, three 95 %, five 99.3 %. Inverting: $t = \tau \ln \dfrac{V_\infty - V_0}{V_\infty - V(t)}$.

2 The two switch rules, and why they hold

THE TWO SWITCH INSTANTS

Capacitor voltage is the integral of a finite current, so it cannot jump: $V_C(0^+) = V_C(0^-)$. Current can jump, because it is set by the resistors.

So at $t = 0^+$: uncharged means a wire, charged means a fixed voltage source. At $t = \infty$ every current has died and the capacitor is an open circuit.

Consequence: a step into a capacitive divider divides by C ; the final state divides by R. That pair of answers is most of what is asked.

DISCHARGE, AND WHY ENERGY DIES TWICE AS FAST

With $E = 0$: $V = V_0 e^{-t/RC}$ and $i = \dfrac{V_0}{R}e^{-t/RC}$.

Energy $U = \frac{1}{2}CV^2 = \frac{1}{2}CV_0^2 e^{-2t/RC}$: the exponent doubles because U goes as V^2.

Settling heat between two voltages: $H = \frac{1}{2}C(V_0 - V_\infty)^2$, whatever R is - smaller R just means a bigger current for a shorter time.

TRAP

At DC steady state the capacitor is open, but its voltage need not be zero. Replacing it by a wire is the $t = 0^+$ picture, not the final one.

3 Reservoir capacitor and the AC response

RESULT

RIPPLE, THEN REACTANCE

Between recharges the capacitor supplies the load: $Q = I\Delta t$, so $\Delta V \approx \dfrac{I_{\text{load}}\Delta t}{C}$. A $1\ \text{A}$ load fed every $10\ \text{ms}$ with $0.5\ \text{V}$ ripple needs $20\ \text{mF}$.

For $V = V_0 \sin \omega t$, $i = C\dfrac{dV}{dt} = \omega C V_0 \cos \omega t$: the current leads by 90° and its size is V_0/X_C with $X_C = \dfrac{1}{\omega C}$.

Series RC : $Z = \sqrt{R^2 + X_C^2}$, $\phi = -\arctan(X_C/R)$; at $X_C = R$ the current leads by 45°.

worked to the last digit

$2\ \mu\text{F}$ at $10\ \text{V}$ discharged through $1\ \text{M}\Omega$: $\tau = 2\ \text{s}$, so at $t = 2\ \text{s}$ the voltage is $10e^{-1} = 3.68\ \text{V}$.

Energy falls from $100\ \mu\text{J}$ to $13.5\ \mu\text{J}$; the missing $86.5\ \mu\text{J}$ is heat in the resistor.

4 The meter changes the reading

KEPT

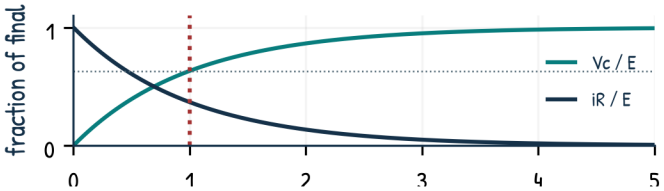
WHY A PROBE LOADS THE CIRCUIT

The meter has its own input capacitance C_m. Connecting it joins two capacitors, so charge is shared.

A capacitor C_s at V_0 then reads $V = V_0 \dfrac{C_s}{C_s + C_m}$: with $C_m = C_s$ you lose half the reading. That is why a small capacitor needs a high-impedance probe.



Everything except the capacitor is one source plus R_{th} : that is why one exponential is enough.



Voltage rises as $1 - e^{-t/\tau}$ while current falls: their areas settle the energy books.

REMEMBER

Solve $RC\dot{V} + V = V_\infty$ once. Then V_0, V_∞, $R_{\text{th}}C$ answer every transient.